Research Overview

My research focuses on how learning systems behave under structural constraints and how those constraints affect adaptation over time. I study sequential models and reinforcement learning agents in settings where common assumptions, such as stable rewards, fixed temporal structure, or Euclidean geometry, do not hold.

In my work on GPT-2 with reinforcement learning, I examined how explicitly defined reward signals shape text generation behavior when applied conservatively to a frozen pretrained model.
In LSTM-based time-series modeling, I analyzed how recurrent models perform under distributional shifts, observing strong performance within the training regime and reduced reliability during regime changes.
In non-Euclidean navigation, I studied how geometric structure influences reinforcement learning behavior, showing partial policy transfer from Euclidean to non-Euclidean environments with increasing degradation as complexity increased.

Together, these projects reflect a consistent interest in understanding how learning algorithms adapt when underlying structural assumptions are violated. This perspective motivates my interest in doctoral research on reinforcement learning and sequential models, with particular attention to robustness and controlled adaptation in non-stationary settings.

Using a GPT-2 Model with Reinforcement Learning for Personalized and Targeted Text Generation

Gogitidze, G. & Veksler, V. D. (2024). Using a GPT with Reinforcement Learning for More Personalized or Targeted Text Generation. *28th annual conference of the Consortium for Computing Sciences in Colleges, Northeastern Region (CCSCNE)*. Albany, NY.

Research Context

Pretrained language models such as GPT-2 generate fluent text but provide limited control over stylistic and qualitative properties such as repetition and diversity. This project investigates whether reinforcement learning can be used to directly optimize generation quality metrics while preserving the linguistic structure of a frozen pretrained model.

Problem Formulation

Text generation was framed as a reinforcement learning problem operating over a fixed GPT-2 policy. At each generation step, the model produced the next token conditioned on the prompt and the partially generated output. The agent's state combined the input prompt with the current text prefix, and actions corresponded to next-token selections from the vocabulary. This formulation enabled direct optimization of output quality rather than likelihood alone.

Methodology

Policy optimization was performed using **Proximal Policy Optimization (PPO)** with a clipped objective to stabilize updates. A small learning rate and limited epochs per batch were used to prevent collapse and preserve fluency during training.

Reward signals were **manually specified** through deterministic heuristics derived from generated text. These rewards were computed automatically at runtime and reflected qualitative properties of the output rather than human annotations. The reward function incorporated:

- penalties for repeated n-gram patterns,
- incentives for higher lexical diversity,
- and length normalization to discourage trivial outputs.

This reward design allowed direct control over generation behavior without introducing a learned reward model or interactive feedback loop.

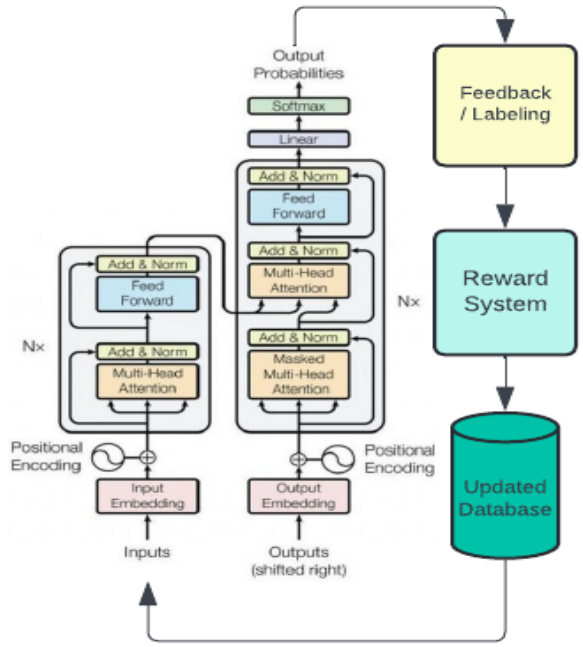


Figure 1. *GPT-2 generation pipeline with specified reward feedback*

Evaluation and Metrics

Evaluation was conducted against a frozen GPT-2 baseline using identical prompt sets. Output quality was assessed using both qualitative inspection and quantitative metrics.

Lexical diversity was measured as the ratio of unique tokens to total tokens across generated outputs, averaged over evaluation prompts. This metric captures variation in word usage and serves as a proxy for repetitive generation behavior.

Results

The reinforcement-learning-optimized model demonstrated consistent improvements over the baseline GPT-2. Lexical diversity increased from **0.499 to 0.616**, indicating more varied word usage across responses. Repetition was reduced by **23%**, reflecting improved control over degenerative generation patterns.

Observed failure modes included reward over-optimization, where excessive emphasis on diversity penalties led to reduced fluency beyond certain thresholds. These cases highlighted the sensitivity of language generation to reward weighting and the need for constrained policy updates.

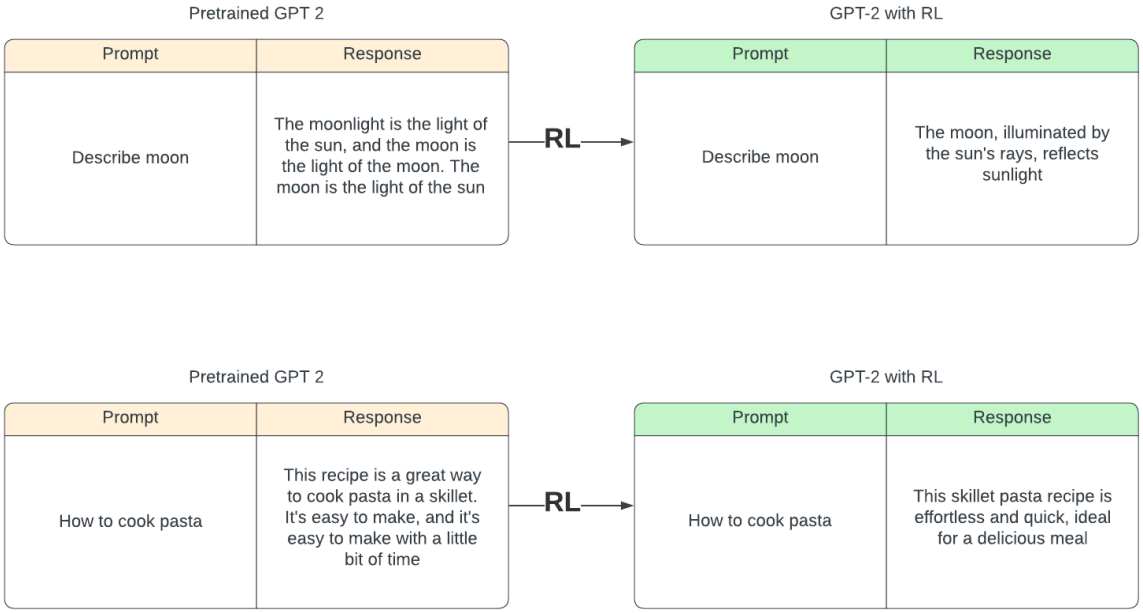


Figure 2. Representative GPT-2 outputs before and after reinforcement learning optimization, illustrating reduced repetition and more concise responses.

Research Insights

This work shows that reinforcement learning can be used to guide pretrained language models toward more controlled and personalized outputs without modifying the underlying architecture. Explicit reward specification proved effective for shaping generation behavior, while PPO provided a stable optimization mechanism when applied conservatively. The findings emphasize the importance of balancing reward design with linguistic coherence in RL-based text generation.

Future Directions

Future work will investigate adaptive learning rates and alternative batching strategies to improve training stability. Extending the approach to more complex prompts and broader datasets may further clarify the scalability of reinforcement learning as a control mechanism for language models.

Machine Learning Model in the Technical Analysis of the Stock Market

Gogitidze, G. & Veksler, V. D. (2023). Machine Learning Model in the Technical Analysis of the Stock Market. *The 6th Annual Caldwell University Research and Creative Arts Day.* Caldwell, NJ.

Research Context

Traditional statistical models commonly used in technical analysis, such as linear regression, often struggle to capture non-linear dynamics present in financial time series. This project explores whether recurrent neural networks, specifically Long Short-Term Memory (LSTM) models, can better model temporal dependencies in equity price data and improve predictive performance under realistic market conditions.

Problem Formulation

The task was framed as a supervised time-series prediction problem using historical equity price data. Given a fixed-length window of past observations, the model was trained to predict future price values. A key challenge addressed in this work is the known difficulty of LSTM models in extrapolating beyond the distribution of the training data, particularly in the presence of regime shifts or extreme market events.

Methodology

Historical stock price data for Microsoft (MSFT) was obtained from the Yahoo Finance database. The raw time-series data was preprocessed using a sliding-window approach, transforming the original dataset into overlapping input–target pairs suitable for recurrent modeling.

An **LSTM network**, a class of recurrent neural networks designed to capture long-range temporal dependencies, was implemented and trained on the processed data. Model performance was evaluated using **mean squared error (MSE)** and **mean absolute error (MAE)**. Training was conducted over multiple epochs to observe convergence behavior and stability.

To mitigate extrapolation issues inherent to LSTMs, the training window was restricted to data beginning on **2021-03-25**, ensuring that the model operated within a regime more representative of the evaluation period.

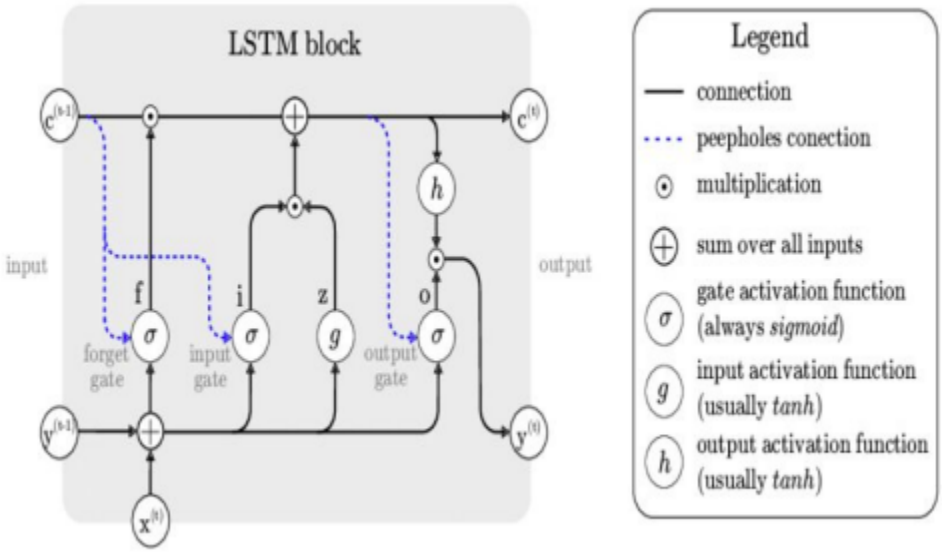


Figure 3. LSTM cell structure illustrates input, forget, and output gates with recurrent state propagation(Figure included to contextualize gating behavior relevant to observed extrapolation failures).

Results

Across training epochs, the model demonstrated stable convergence behavior. Mean absolute error decreased from **3.6484 at epoch 100** to **3.0887 at epoch 500**, indicating improved predictive accuracy with continued training. The LSTM model outperformed a baseline linear regression approach on the same task, confirming its ability to capture non-linear temporal patterns in stock price movements.

Limitations and Observations

Despite improved performance within the training distribution, the LSTM model exhibited limited effectiveness when extrapolating beyond the observed data range. This behavior is consistent with known limitations of recurrent models when faced with distributional shifts or rare market events. These findings highlight the importance of careful data selection and the inherent trade-offs between flexibility and generalization in time-series modeling.

Research Insights

This project demonstrates that LSTM models can provide meaningful improvements over traditional linear methods for financial time-series prediction when operating within a stable data regime. At the same time, it underscores the challenges of extrapolation in neural sequence models and motivates further investigation into hybrid approaches that combine recurrent architectures with models better suited for handling regime changes.

Future Directions

Potential extensions include incorporating additional input features such as moving averages and fundamental indicators, as well as exploring alternative machine learning approaches such as gradient-boosted decision trees. These directions aim to improve robustness and predictive accuracy, particularly under volatile or non-stationary market conditions.

Non-Euclidean Navigation in Reinforcement Learning

Summer Paid Research
Jun 2022 – Aug 2022

Research Context and Motivation

Most navigation algorithms and learning environments implicitly assume **Euclidean geometry**, where distances, angles, and parallelism follow flat metric properties. While this assumption is practical for many real-world tasks, it fails to capture settings where curvature plays a fundamental role, such as hyperbolic or spherical spaces. Relaxing Euclidean assumptions has

historically led to advances in mathematics, physics, and our understanding of spatial structure, particularly in contexts involving curved manifolds.

Advances in **virtual reality and game engines** enable the construction and exploration of non-Euclidean environments that would be difficult or impossible to realize physically. Recent non-Euclidean virtual environments and games have demonstrated how curvature fundamentally alters navigation, perception, and spatial intuition, motivating their use as experimental platforms for studying learning agents under geometric distortion. This project leverages such environments to study how reinforcement learning agents adapt to navigation tasks when the underlying space violates Euclidean assumptions.

Environment Design

A non-Euclidean navigation environment was constructed in **Unity**, based on **hyperbolic geometry** derived from a **hyperbolic paraboloid** surface. The environment emphasized defining properties of non-Euclidean space, including divergent parallel lines and deviations from Euclidean triangle angle relationships. These geometric features altered distance and direction cues available to the agent while maintaining a controlled navigation objective.

An agent and a target object were placed within the environment, with randomized initial positions at the beginning of each episode. Explicit environmental boundaries were enforced to constrain exploration and induce meaningful navigation behavior under curved spatial structure.

State, Action, and Reward Specification

The agent's state representation consisted of its position, the target's position, and relative target vectors. A **discrete action space** was used to facilitate analysis of navigation behavior under geometric constraints.

Reward shaping employed **distance-based incentives** for movement toward the target and penalties for undesirable behaviors such as boundary violations. Episodes terminated upon reaching the target or violating predefined constraints. This formulation was designed to isolate the effect of geometry on learned navigation policies rather than rely on task-specific heuristics.

Learning Algorithm

Agents were trained using a **modified Delayed Q-learning algorithm with PAC guarantees**, selected for its stability properties under altered transition dynamics. The learning setup emphasized controlled updates and robustness, allowing examination of policy behavior as geometric complexity increased.

Observations and Findings

Learned navigation policies exhibited **partial generalization** when transitioning from Euclidean to non-Euclidean state spaces. As environment complexity and geometric distortion increased, performance degradation was observed. This behavior aligns with expectations regarding representation mismatch and highlights the sensitivity of learned policies to underlying spatial structure.

Research Insights

This work demonstrates that reinforcement learning agents implicitly rely on Euclidean assumptions embedded in their state representations and environment dynamics. When these assumptions are violated, navigation performance degrades in predictable ways. Studying navigation in non-Euclidean environments provides a controlled setting for examining how geometry influences learning and motivates future research into geometry-aware representations and learning methods capable of operating across diverse spatial manifolds.

Appendix A
CCSCNE 2024 Research Poster

Using a GPT with Reinforcement Learning for More Personalized or Targeted Text Generation
Giorgi Gogitidze, Vladislav D. Veksler
28th Annual Conference of the Consortium for Computing Sciences in Colleges, Northeastern Region (2024)

(The following page reproduces the full conference poster.)

Using a GPT with Reinforcement Learning for More Personalized or Targeted Text Generation

CALDWELL UNIVERSITY
School of Business and Computer Science

Giorgi Gogitidze (ggogitidze@caldwell.edu)
Vladislav D. Veksler (vveksler@caldwell.edu)

Problem

- Conventional GPT models, though effective, lack adaptability for personalized or context-targeted responses, resulting in generic outputs despite user feedback.
- Users may receive less precise and contextually relevant text due to the models' limited adaptability to individual criteria.
- The absence of personalized adaptation mechanisms in GPT models hinders the fulfillment of specific user needs and preferences.

Pretrained GPT 2

Prompt	Response
Describe moon	The moonlight is the light of the sun, and the moon is the light of the moon. The moon is the light of the sun

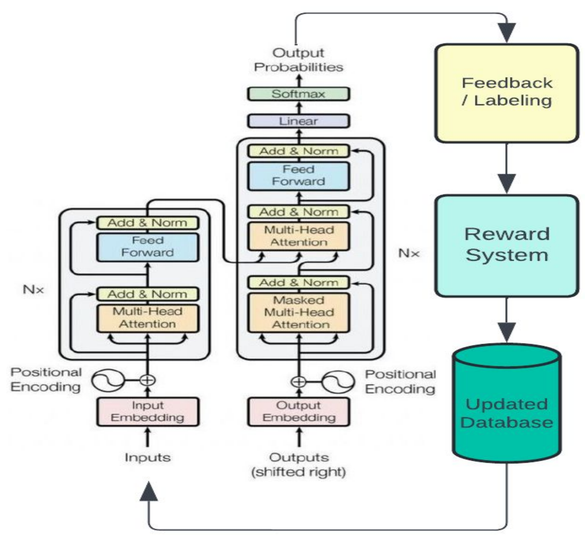
Pretrained GPT 2

Prompt	Response
How to cook pasta	This recipe is a great way to cook pasta in a skillet. It's easy to make, and it's easy to make with a little bit of time

Solution

Our research explores enhancing GPT-2 with Reinforcement Learning (RL) to refine text output towards more personalized and context-specific responses.

- By manually assigning rewards to outputs based on quality, we directly influence and optimize the learning process, minimizing unnecessary computational efforts.



Importance

- Enhanced adaptability and personalization lead to more tailored responses.
- Specific contextual considerations improve the relevance and effectiveness of generated text.
- Diverse applications benefit from the expanded capabilities of text-generation models.

Methodology

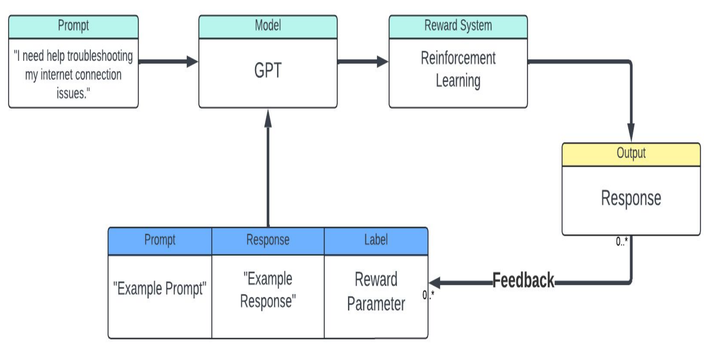
We used GPT-2 [1] with manual reward assignment based on the quality of generated text.

- Directly influencing the model's learning process by providing immediate feedback for each output.
- Enhanced the efficiency of the training phase and tailoring the GPT-2 model's responses to desired criteria without extensive computational overhead.

```
with tf.GradientTape() as tape:
    input_ids = tokenizer.encode_batch(batch_prompts, return_tensors="tf")
    outputs = model(input_ids, labels=input_ids)
    losses = [-(reward / 2) * output.loss for reward, output in zip(rewards, outputs)]
    total_loss = tf.reduce_mean(losses)

gradients = tape.gradient(total_loss, model.trainable_variables)
optimizer.apply_gradients(zip(gradients, model.trainable_variables))

loss_scalar = total_loss.numpy().item()
```



Results

Qualitative text analysis of the proposed approach revealed more precise and concise text generation capabilities.

- we observed a reduced number of repetitions in text responses. For original GPT-2 mean lexical diversity was 0.499, while for GPT-2 with RL showed lexical diversity of 0.616.
- These findings suggest that the proposed method has the potential to augment flexibility and precision of pre-trained language models across diverse applications.

Prompts	GPT2 responses	GPT2 with RL responses
1. How to start a completely new career in period of two years?	I think it's a good idea to start a new career in period of two years.	Starting a new career in two years is smart.
2. I want to know When is the best timing to start a business this year?	The best timing is when you have a good product or service.	Start when your product or service is ready.
3. What is the best way to save money for a down payment on a house?	The best way to save money for a down payment on a house is to buy a house. If you are looking for a house that is priced right.	Save by setting aside money and finding a good price.
4. Explain the concept of neural networks and how they work.	The first step is to understand how neural networks work. The second step is to understand how they work.	Neural networks learn patterns for predictions.
5. What are the key differences between a Roth IRA and a traditional	A Roth IRA is a traditional IRA that is a Roth IRA	Roth IRA and traditional differ in tax handling.

Conclusion

Integrating GPT with RL offers a promising avenue for enhancing text generation models' efficiency and personalization.

- Our findings suggest significant improvements in model responsiveness and relevance.
- It explores potentially more natural and personalized AI-driven text interactions.

Future research will involve refining proposed optimization techniques and evaluating its applicability on more complex generation tasks.

- Will explore the potential for incorporating adaptive learning rates and exploring the effects of different batching strategies on model performance and resource utilization.

References

https://huggingface.co/openai-community/gpt2

https://huggingface.co/datasets/Dahoas/sft-gptj-synthetic-prompt-responses

https://arxiv.org/abs/1706.03762

Appendix B Caldwell University Research & Creative Arts Day Poster

Title: Machine Learning Model in the Technical Analysis of the Stock Market

Author(s): Giorgi Gogitidze& Vladislav D. Veksler

Venue: Caldwell University Research & Creative Arts Day, 2023

(The following page reproduces the full research poster associated with this project.)

Machine Learning Model in the Technical Analysis of the Stock Market

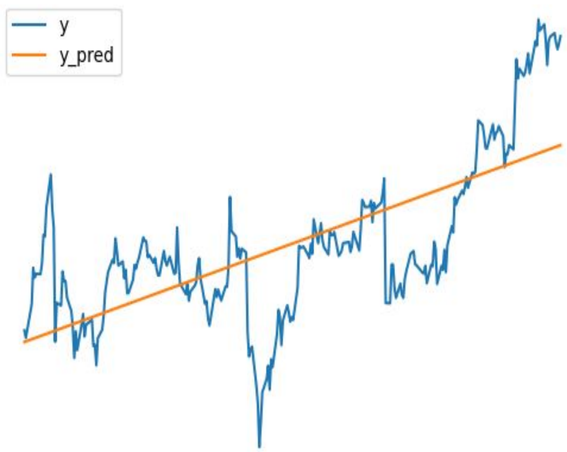
Giorgi Gogitidze (ggogitidze@caldwell.edu) & **Vladislav D. Veksler** (vveksler@caldwell.edu)

CALDWELL UNIVERSITY

Problem

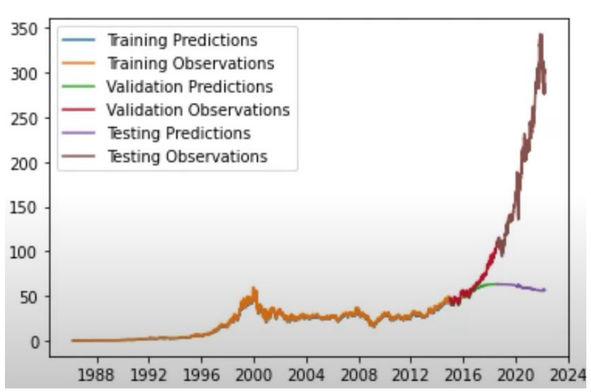
Traditional statistical models used in technical analysis, such as linear regression, may be limited in their ability to account for these non-linearities.

- Can machine learning algorithms, such as LSTM models, overcome this limitation and improve the accuracy of technical analysis?



LSTM models are generally poor at extrapolating beyond the range of the training data.

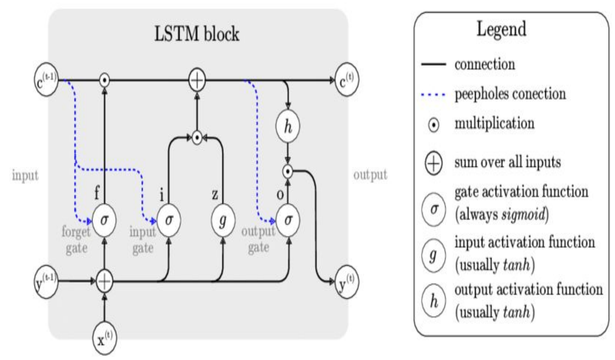
- LSTM models are designed to model patterns in sequential data, and when the input data falls outside the range of the training data, the model may not be able to accurately predict the output



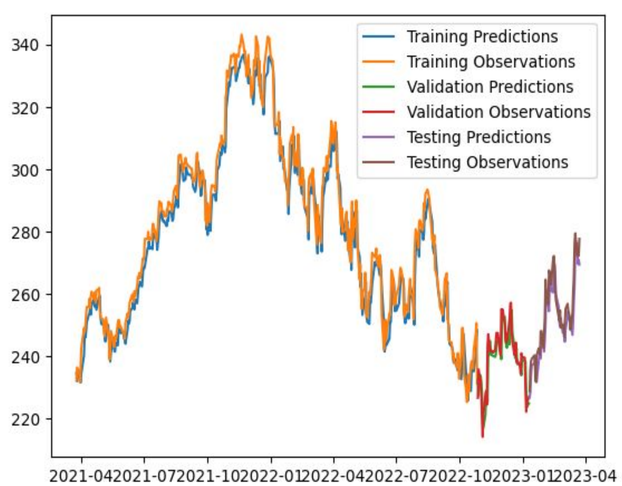
Solution

LSTM(Long-Short Term Memory) model, which is a type of RNN(Recurrent Neural Network)is implemented for technical analysis in the stock market.

- Well-suited for time series data.
- Able to capture non-linear patterns in the data and adapt to changing market conditions.
- Designed to work with sequential data and can identify important patterns and trends.



LSTM is poor at extrapolating (learning data outside its range), so we changed the starting data to **2021-03-25** so that model will be able to calculate sequence.



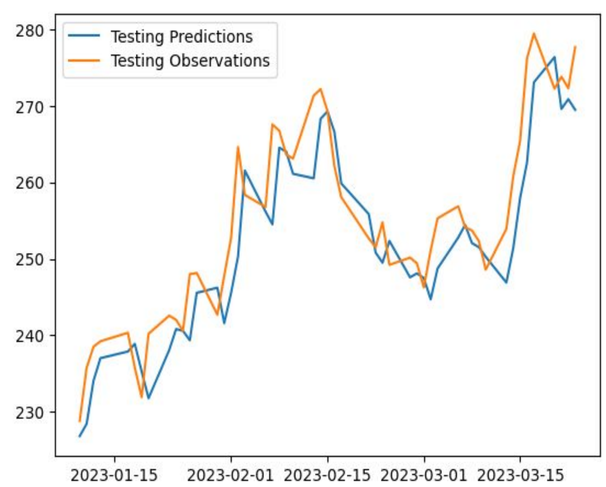
Methodology

- Data acquisition:obtained historical stock price data for Microsoft (MSFT) from the Yahoo Finance database.
- Data preprocessing: we changed the original DataFrame by converting it into a new DataFrame with sliding windows.

	Target Date	Target-3	Target-2	Target-1	Target
0	2021-03-25	235.990005	237.580002	235.460007	232.339996
1	2021-03-26	237.580002	235.460007	232.339996	236.479996
2	2021-03-29	235.460007	232.339996	236.479996	235.240005
3	2021-03-30	232.339996	236.479996	235.240005	231.850006
4	2021-03-31	236.479996	235.240005	231.850006	235.770004
...	...	...	...	...	...
498	2023-03-17	260.790009	265.440002	276.200012	279.429993
499	2023-03-20	265.440002	276.200012	279.429993	272.230011
500	2023-03-21	276.200012	279.429993	272.230011	273.779999
501	2023-03-22	279.429993	272.230011	273.779999	272.290009
502	2023-03-23	272.230011	273.779999	272.290009	277.660004

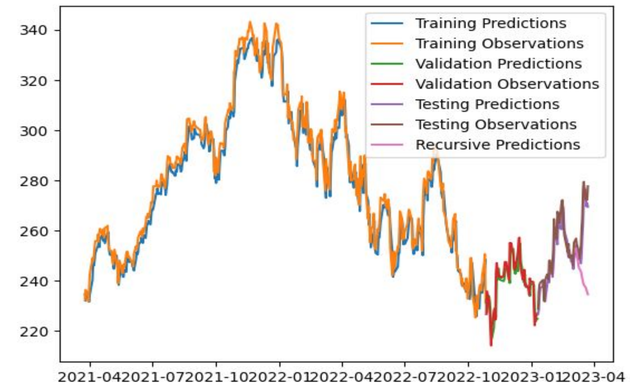
503 rows × 5 columns

- Evaluation: We evaluated the performance of the LSTM model by calculating the mean squared error (MSE) and mean absolute error.



Results

Epoch **100** mean_absolute_error: **3.6484**
Epoch **200** mean_absolute_error: **3.1332**
Epoch **300** mean_absolute_error: **3.5162**
Epoch **400** mean_absolute_error: **3.1482**
Epoch **500** mean_absolute_error: **3.0887**



Conclusion

LSTM outperformed traditional linear regression models in predicting stock prices. LSTM is not as effective at extrapolating beyond the training data, highlighting the need for continued research in developing models that can better handle extreme or unexpected events.

Future Research

- Adding input variables such as moving averages and fundamental analysis to improve the accuracy.
- Using Gradient Boosted Decision Trees as alternative machine learning approach